



Tuning direct CO hydrogenation reaction over Fe-Mn bimetallic catalysts toward light olefins: Effects of Mn promotion

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ABSTRACT

Highly dispersed FeMn bimetallic nanoparticles prepared by a low-temperature co-precipitation method were employed for direct CO hydrogenation for light olefins ($C_2\sim C_4$). An outstanding selectivity (60.6 %) for light olefins at a CO conversion level of 7.6 % was obtained over FeMn (4:1) catalyst activated by syngas (5%CO/5% H_2). Characterization indicated that a strong interaction between Fe and Mn resulted in significant effects on stabilization of $Fe_{1-x}Mn_xO$ phase during activation period. Responding to the morphology effect, Mn decreased the surface energy of nanocrystals and weakened the agglomeration of nanoparticles, thus resulted in a decrease in particle size of Mn-promoted catalysts. The combined effects of inhibited carburization and improved metal dispersion contributed to the volcano pattern in the catalytic activity of Mn-promoted catalysts. In addition, Mn may act as an effective electron donor to Fe, enhancing CO dissociative adsorption while inhibiting secondary hydrogenation of olefins, thus giving rise to high olefin selectivity in CO hydrogenation.

1. Introduction

Light olefins, namely $C_2\sim C_4$ olefins, play important roles in chemical industry because they are building blocks of advanced chemicals and materials such as plastics, synthetic textiles, lubricants, detergents, solvents and cosmetics [1]. Currently, light olefins are primarily produced from steam cracking of a broad range of petroleum feedstocks, ranging from ethane to naphtha and gas oil. Although steam cracking is a well-established, large-scale technology, the constantly growing demand for light olefins, the high energy-consumption and large CO_2 -footprint of steam cracking, as well as the fluctuating availability and price of crude oil motivate the development of alternative production technologies starting from non-petroleum feedstocks, such as coal, biomass, CO_2 , and natural gas. Countries with large natural gas or coal reserves have pioneered the research and development of these alternative processes to produce light olefins to reduce their dependence on oil imports. Among those countries, China represents the largest market in the world for Coal-to-Olefins, which mainly consists of four stages, namely, (i) syngas production by coal gasification, (ii) syngas conditioning, (iii) synthesis of olefins, and (iv) product separation [2].

Generally, synthesis of olefins from syngas can be realized via either indirect or direct route. In the indirect route, olefins are produced via the synthesis of intermediates such as methanol or dimethyl ether from syngas, followed by methanol/dimethyl ether-to-olefins process (MTO/DMTO). Syngas can also be converted directly into light olefins via a technology derived from Fischer-Tropsch Synthesis (FTS), the so-called Fischer-Tropsch to Olefins (FTO) process, or a novel bifunctional catalyst technology proposed by Bao's and Wang's groups [3,4]. In recent years, a lot of work has emerged through the regulation of the types of oxides and molecular sieves, as well as the intimacy of dual active sites [5,6]. Compared with the indirect approach, the direct route has the advantages of process simplicity, lower energy consumption and lower investment cost. However, despite receiving considerable attention nowadays, the dual-function catalysts with superior light olefin selectivity suffer from low activity and carbon utilization efficiency [7]. Although the FTO process has gained increasing attentions in recent decades, commercialization of this promising technology is hampered by the limited $C_2\sim C_4$ olefin selectivity, the inherent formation of undesirable byproducts (i.e., CH_4 and CO_2), and the poor stability of existing catalyst systems [3].

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Cobalt and iron are the most common FTS catalyst materials. Fe-based catalysts have a significant price advantage over Co-based catalysts. In addition, Fe-based catalysts show (1) high water gas shift (WGS) activity, which is favorable for the conversion of CO-rich syngas derived from biomass or coal; (2) higher olefins selectivity due to their limited hydrogenation activity; and (3) a higher tolerance to poisoning by sulfur, a major contaminant in syngas [8]. These benefits make Fe-based catalysts attractive candidates for the FTO process.

During catalyst activation and reaction, different iron phases, including metallic iron (Fe^0), iron carbides (FeC_x), and iron oxides (Fe_2O_3 and Fe_3O_4) are formed and coexist. Until now, the nature of the active phase and the interplay between different phases has not been elucidated. It is widely recognized that the active phase is related to the formation of surface iron carbides. Cementite ($\theta\text{-Fe}_3\text{C}$), Hagg carbide ($\chi\text{-Fe}_5\text{C}_2$) and Fe_7C_3 are three bulk carbides which are stable at FTS conditions [9]. However, these carbides are prone to undergo phase transitions when they are exposed to air due to their pyrogenic nature [10], which makes their characterization challenging. Other studies [11, 12] have reported that phase transitions of iron carbides can also be driven by temperature changes. The air and temperature sensitivity of iron carbides creates challenges for characterizing the active phases during FTS since phase transitions may occur during pretreatment, during reaction and during analysis. Consequently, the current lack of detailed understanding of the formation and evolution of active phases (both carbides and oxides) during activation and reaction, and the effect of promoters and reaction conditions on these transformations seriously limit rational catalyst design.

It is well known that the intrinsic activity and selectivity of a catalyst are closely linked to its electronic properties and the geometric structure, which can be modified with the introduction of promoters. Manganese (Mn) oxides have been frequently used as promoters to improve light olefin selectivity, activity, and stability of iron-based FTO catalysts [7,13]. However, the promotion function of Mn is still not fully understood as Mn doping shows a strong interplay of structural and electronic effects in iron-based FTO catalysts. Mn is preferred to be considered as a structural promoter that improves the dispersion of active sites by changing the structure of the active phase and thus increase the intrinsic activity [14]. For example, Mn atom can incorporate into the lattice of iron oxides on the catalyst surface and form a solid solution of Mn during the processes of calcination [15,16]. Or, Mn oxides can migrate to the catalyst surface and form an Mn enriched layer during the calcination and activation processes. Our previous study on spinel-type Mn-doped ferrites also observed the enrichment of MnO_x on the catalyst surface [17]. Besides, Mn can act as an electronic promoter that effectively reduces the hydrogenation probability of $\text{C}=\text{C}$ and thereby leads to a high selectivity towards light olefins, owing to its strong surface basicity. Earlier studies found that Mn promoter played an essential role in the formation of different surface iron carbides, which dictated product selectivity in FTO [14,18]. In addition, the suppression effect of Mn on the carbonization of iron oxides due to the strong interaction between Fe and Mn was also reported [19]. From literature survey, there has been disagreement over the role of Mn in Fe-Mn system. Such disagreement is not only caused by different catalyst synthesis methods, but also subjected to the changes in activation and reaction conditions. Therefore, further investigation is needed to unravel the intrinsic role of Mn promoter in Fe-Mn FTO system.

Herein, we synthesized a series of highly dispersed Fe-Mn bimetallic oxides with similar sizes using a low-temperature (-10°C) coprecipitation method. The activity and light olefin selectivity exhibited a volcano-type curve against the Fe/Mn ratio in the catalysts. An outstanding light olefin selectivity (60.6 %) at a CO conversion of 7.6 % was achieved by activating the FeMn bimetallic catalyst with the optimum Fe/Mn ratio using syngas (5%CO/5%H₂/90 %Ar). The phase transitions of the iron oxides in the as-prepared FeMn bimetallic catalysts during activation and reaction were characterized using multiple ex situ/in situ techniques, including transmission electron microscopy (TEM), X-ray

photoelectron spectroscopy (XPS), ex/in situ X-ray diffraction (XRD), and extended X-ray absorption fine structure (EXAFS). By correlating the information on the nature of the iron sites during different stages of FTO reaction with catalytic activity and selectivity, we proposed the promoting effects of Mn in FeMn bimetallic FTO catalysts.

2. Materials and methods

2.1. Catalyst synthesis

Industrial FTS catalysts are typically prepared using impregnation or precipitation methods. Compared to the traditional high temperature/room temperature precipitation method, low temperature precipitation favors the formation of crystal nuclei, accelerates the nucleation and splitting of crystals, and finally contributes to nano-catalytic materials with high surface energy and specific surface area [20]. The FeMn bimetallic oxides were prepared via a low temperature coprecipitation method as described elsewhere [20]. Briefly, a mixture containing $\text{FeCl}_2\cdot 4\text{H}_2\text{O}$ (AR, Sinopharm) and $\text{MnCl}_2\cdot 4\text{H}_2\text{O}$ (AR, Sinopharm) with a fixed total metal ion of 0.1 mol was dissolved in 30 mL ethylene glycol under vigorous stirring in N_2 protection. The solution was cooled to -10°C and maintained under this temperature for 10 min. An aqueous Na_2CO_3 solution (100 mL, 0.2 M) was slowly added at a rate of 1.2 mL/min. Then, the mixture was aged for 1 h, to recover the precipitate, which was then washed with deionized water and ethanol, and finally dried at 60°C overnight. The FeMn oxides were obtained by calcinating the solid at 450°C for 4 h. The as-prepared FeMn oxides were denoted as $\text{FeMn}(x:1)$, where x is the nominal atomic ratio of Fe:Mn. For comparison, monometallic oxides, iron oxide (Fe_2O_3) and manganese (IV) oxide (MnO_2), prepared by low-temperature precipitation were used as reference catalysts.

2.2. Catalyst characterization

Textural properties of the catalysts were measured by N_2 physisorption at -196°C using a Micromeritics ASAP-2010 system. The samples were degassed under vacuum at 200°C for 6 h. The specific surface area was calculated by Brunauer-Emmett-Teller (BET) method. The actual metal contents of Fe and Mn in the as-prepared catalysts were determined by inductive coupled plasma mass spectrometry (ICP-MS, Perkin Elmer, NexION 300). The surface morphology and particle size of the catalysts were obtained using a high-resolution transmission electron microscope (HRTEM, JEOL, JEM-2100). Prior to imaging, the catalyst sample was ultrasonically dispersed in ethanol and dropped onto a carbon-coated copper grid. Diffuse reflectance ultraviolet-visible (UV-vis) spectra were collected to reveal the chemical states of the catalyst on an UV-vis spectrophotometer (Lambda 750, PerkinElmer, Inc.) in a region of wavenumber range 200–800 nm. Powder XRD was performed with a Bruker D8 advance diffractometer using $\text{Cu-K}\alpha$ radiation ($\lambda = 1.5406 \text{ \AA}$) with an accelerated voltage of 40 kV. The XRD patterns of fresh and spent catalysts were recorded with a step of 0.02° and scanning rate of $0.24^\circ/\text{min}$ in a scanning angle range (2θ) of 10° – 80° . The average crystallite size of the fresh and spent catalysts was calculated from the full width of half maximums (FWHM) of major crystalline facets at (3 1 1), (4 4 0) for Fe_3O_4 , (1 1 1), (2 0 0) for FeO , (1 1 1) for Fe_2C and (5 1 0) for Fe_5C_2 by Scherrer's equation Eq. (1).

$$D = \frac{0.89\lambda}{\Delta \cos \theta} \quad (1)$$

Where D is the crystal size, and λ , Δ and θ are wavelength of X-rays, FWHM of diffraction peaks and the angle of the corresponding peak. Rietveld refinement of the diffraction patterns were performed using TOPAS package. In situ XRD patterns of the as-prepared catalysts during activation were recorded using the same instrument coupled with a reaction chamber. The catalyst samples were reduced in 5% H_2/Ar (50

mL/min) from room temperature to 350 °C with a ramping rate of 5 °C/min and then held at that temperature for 5 h. The XRD patterns were collected at specified temperatures with a scanning rate of 10°/min. ⁵⁷Fe Mössbauer spectra were collected at room temperature using a WissEL/MSPCA spectrometer irradiated with a ⁵⁷Co source. The spectra were recorded in a constant transmission mode using a proportional counter. Peak fitting was conducted for phase composition analysis.

XPS spectra were obtained with a PHI 5000 Versaprobe spectrometer using a monochromatic Al-K radiation source (1486.6 eV). The binding energies (B.E.) were calibrated using C 1s adjusted to 284.6 eV. For XPS measurements of the spent catalysts, the catalysts were passivated in 1.0 %O₂/N₂ for 5 min and immediately transferred to the XPS chamber. Prior to XPS measurements, the passivated layers were removed by in situ argon ion (Ar⁺) sputtering with a beam energy of 2 kV for 2 min. The X-ray absorption spectra at Fe K-edge and Mn K-edge were obtained in the transmission mode at the BL14W1 beamline of the Shanghai Synchrotron Radiation Facility (SSRF) which was operated at an electron beam energy of 3.5 GeV. Fe oil, FeO, Fe₃O₄, Fe₂O₃, γ-Fe₂O₃, Mn foil, MnO, MnO₂, Mn₂O₃ and Mn₃O₄ were used as references. X-ray absorption near-edge structure (XANES) and extended X-ray absorption fine structure (EXAFS) analysis were performed using Athena and Artemis software package.

H₂ temperature programmed reduction (H₂-TPR) experiments were carried out in a micro fixed-bed reactor coupled with a thermal conductivity detector (TCD) under 10 % H₂/Ar flow (50 mL/min) in the range of 30–800 °C with a ramping rate of 10 °C/min. Temperature programmed surface reaction in CO/H₂ mixture (CO/H₂-TPSR) was carried out in the same reactor as H₂-TPR. The catalysts were in situ reduced in the reactor at 350 °C for 5 h. After cooling to the room temperature under Ar purging, a syngas mixture (5%CO/5%H₂/Ar, 30 mL/min) was introduced into the reactor. The surface reaction was taken in the temperature range of 30–800 °C at a heating rate of 10 °C/min. The desorbed products (CH₄, CO₂, C₂H₄, and H₂O) were monitored by a mass spectroscopy detector (MSD, Hiden HPR 20).

2.3. Catalytic performance

The catalytic performance in FTO was evaluated in a stainless-steel fixed bed reactor (Length: 450 mm, i.d.: 9 mm). More details on the experimental setup and reaction procedure are available in our previous work [17,21]. Briefly, 25 mg catalyst diluted with 75 mg SiC was activated at 350 °C for 5 h and then tested under 260 °C, 2.0 MPa, H₂/CO = 1, and gas hourly space velocity (GHSV) = 4000 h⁻¹. CO conversion, product selectivity, space-time yield (STY) of C₂-C₄ olefins and iron time yield (FTY) denoted as moles of CO converted per gram of iron per second, were calculated using Eq. (2) ~Eq. (5).

$$X_{CO}(\%) = \frac{Q_{out} \sum v_i C_i}{Q_{in} v_{CO}} \times 100\% \quad (2)$$

Where Q_{in} and Q_{out} are the volume flow rate of inlet and outlet stream in mL/min. v_i and v_{CO} are the volume fraction of component i in the outlet stream and volume fraction of CO in the feed stream. C_i is the carbon number of component i .

$$S_i(\%) = \frac{Q_{out} v_i C_i}{Q_{in} v_{CO} X_{CO}} \times 100\% \quad (3)$$

The STY of C₂-C₄ olefins ($STY_{C_{2-4}}$) was calculated using:

$$STY_{C_{2-4}} (g \cdot kg_{cat}^{-1} \cdot h^{-1}) = \frac{Q_{out} \sum_{i=2}^4 v_i M_i}{V_m m_{cat}} \quad (4)$$

$$FTY (mol \cdot g_{Fe}^{-1} \cdot s^{-1}) = \frac{Q_{in} v_{CO} X_{CO}}{V_m m_{cat} \sigma} \quad (5)$$

Where M_i is the molecular weight of component i , m_{cat} is the mass of the

catalyst, V_m is the molar volume of an ideal gas at standard temperature and pressure ($V_m = 22,400$ mL/mol), and σ is the mass fraction of iron in the catalyst.

The chain growth probability α , which describes the ratio of chain growth and chain termination is determined by solving the Anderson-Schulz-Flory (ASF) distribution equation (Eq. 6)

$$\ln\left(\frac{W_n}{n}\right) = n \ln \alpha + \ln \frac{(1-\alpha)^2}{\alpha} \quad (6)$$

Where, W_n is the mass fraction of hydrocarbon compound with n carbon atoms.

3. Results

3.1. Structure of the as-prepared Fe-Mn catalysts

The physical properties and structure of Fe-Mn bimetallic catalysts were analyzed using BET, XRD, and TEM. As shown in Fig. 1, the diffraction peaks at 24.1°, 33.2°, 35.6°, 40.8°, 49.5°, 54.1°, 62.4° and 64.0° are assigned to rhombohedral structure of α-Fe₂O₃ (JCPDS No. 86-0550), and the peaks at 30.2°, 35.6°, 43.3°, 53.7°, 57.3° and 62.9° are assigned to cubic structure of γ-Fe₂O₃ (JCPDS No. 39-1346), manifesting that pure iron catalyst is dominated by α-Fe₂O₃ and γ-Fe₂O₃ phases. Since γ-Fe₂O₃ has the same diffraction patterns with that of Fe₃O₄, the presence of gamma phase is further confirmed by the presence of UV-vis characteristic features at 655 nm (Fig. S1). The gamma phase is not visible in Fe-Mn bimetallic catalysts, indicating that the addition of Mn inhibits the nucleation and growth of γ-Fe₂O₃ crystals. As the Fe/Mn ratio decreases from 10 to 2, the diffraction peaks become wider and less intensive, indicating that the crystallinity of the Fe-Mn bimetallic catalysts decreased with the increasing Mn doping amount. Further increasing Mn amount to Fe/Mn ration of 1 results in the disappearance of α-Fe₂O₃ characteristic peaks at high angles. Moreover, the diffraction pattern of FeMn(1:1) slightly shifts to the lower angles (e.g., 23.7°, 32.3°, 35.6°, 40.6°, 48.9° and 53.2°) which are corresponding to Mn₂O₃, manifesting that iron atoms can be incorporated into the framework of the Mn₂O₃ crystals. MnO₂ is the only phase detected in the as-prepared Mn catalyst. From UV-vis spectra (Fig. S1), the monometallic Fe catalyst exhibits three major bands in the region of 250–400 nm, 480–580 nm and 655 nm, where the peak in the UV region is ascribed to the ligand to metal charge transfer (LMCT) between Fe species and oxygen lattice [22], the peak in the region of 400–600 nm corresponds to the photo excited electron pair transition of Fe³⁺, and the shoulder peak near 640 nm can be ascribed to the ⁶A₁-⁴T₂ transition of octahedrally coordinated Fe³⁺, which is featured in γ-Fe₂O₃. No signals of γ-Fe₂O₃ were observed for Fe-Mn bimetallic catalysts, which agrees with the XRD results. The decrease in the peak intensity upon increasing Mn doping amount points to the incorporation of Mn atoms into the structure of α-Fe₂O₃.

The textual information of the as-prepared Fe-Mn bimetallic catalysts is summarized in Table 1. ICP-MS analysis reveals that the actual

Table 1
Composition, specific surface area and the average crystallite size of the Fe-Mn catalysts.

Catalyst	Actual Fe/Mn molar ratio ^a	S _{BET} (m ² /g)	Average crystallite size (nm) ^b
Fe	N.A.	25	19.6
FeMn (10:1)	10.86	57	18.2
FeMn(4:1)	4.73	88	16.1
FeMn(2:1)	1.92	83	11.4
FeMn(1:1)	1.09	90	4.4
Mn	N.A.	69	6.2

^a“N.A.” means not applicable.

^a Determined by ICP-MS.

^b Estimated based on Scherrer equation.

Fe/Mn molar ratio matches to the nominal value, indicating that the catalyst composition can be accurately adjusted via the low temperature precipitation method. The specific surface area of the as-prepared catalysts shows an obvious increasing trend from 25 to 90 m²/g with the increasing Mn content from 0–50%, whereas the crystallite size decreases from 19.6 to 4.4 nm accordingly. Thus, the addition of Mn is confirmed to improve the dispersion of iron oxide, which is consistent with the studies on Mn-promoted iron catalyst reported in literature [17, 23].

TEM observations (Fig. S2) reveal that the as-prepared monometallic Fe catalyst consists of nanosphere with an average diameter of 20–40 nm. The FeMn(4:1) catalyst presents mostly ellipsoidal morphology with a smaller crystallite size (10–30 nm) than monometallic Fe catalyst. Further increase in Mn content to 50 % causes dramatic decrease in the crystallinity of the catalyst, which yields an amorphous morphology in TEM image (Fig. S2 (e, f)). EDS mapping demonstrates that Fe and Mn elements are evenly distributed in the as-prepared catalysts (Fig. 2). The observations from high-resolution TEM (HRTEM) further reveal the distinctiveness of the as-prepared catalysts. It is seen that both monometallic Fe and FeMn(4:1) catalysts consist of highly ordered crystals. The measured lattice spacings of 0.27 nm, 0.25 nm, and 0.37 nm (Fig. 3 (a, b)) are related to the (104) plane of α -Fe₂O₃, the (311) plane of γ -Fe₂O₃, and the (012) plane of α -Fe₂O₃/Mn₂O₃, respectively. In the case of FeMn(1:1), the orientations of the crystal planes are less ordered compared to those of monometallic Fe and FeMn(4:1) catalysts. A MnO₂ (101) layer with a measured lattice spacing of 0.21 nm is identified on the edge of the lattice fringe. In addition, the (110) plane of Mn₂O₃ with a lattice spacing of 0.25 nm is observed inside the nanosphere. These results suggest that Mn doping in small quantity can change the orientation characteristics of the crystal plane of the catalysts, while excess Mn doping will lead to significant lattice distortion, thus decreasing the crystallinity of the Fe-Mn catalysts.

X-ray absorption spectroscopy is a highly sensitive technique to gain information about the oxidative state and coordination of the metal species in the catalysts. We have obtained X-ray absorption near-edge structure (XANES) spectra of the Fe *K*-edge and Mn *K*-edge for monometallic Fe, FeMn(4:1), FeMn(1:1) and monometallic Mn catalysts. As seen in Fig. 4(a), both monometallic Fe catalyst and Fe-Mn bimetallic catalysts show a sharp pre-edge peak (1 s→3d transition) at 7113 eV, an absorption edge (1 s→4p transition) at 7123 eV and a white line at 7133 eV, which are typical characteristics of iron oxide (III) [24]. The Fourier-transformed (*k*³-weighted) EXAFS spectra of the catalysts (Fig. 4 (c)) show three distinct scattering peaks at 1.45 Å, 2.57 Å and 3.21 Å, which closely resemble those of bulk Fe₂O₃ standard. The first peak is

attributed to the Fe-O shell coordination, while the second and the third peak are mainly attributed to the Fe-Fe shell coordination [25]. It is also noted that the intensity of the last two peaks is significantly reduced in Mn doped catalysts, suggesting that Mn induced the formation of disordered structures in Fe₂O₃.

Fig. 4(d)–(f) shows the XANES spectra for the as-prepared monometallic Mn and Fe-Mn bimetallic catalysts, together with the XANES spectra of Mn₃O₄, Mn₂O₃, MnO₂, MnO and Mn foil as the reference materials. The two Fe-Mn bimetallic catalysts exhibit a pre-edge peak at 6541 eV and an absorption edge at 6539 eV, corresponding to the characteristic of Mn (III). In comparison, the monometallic Mn catalyst shows an absorption edge at 6552 eV, which is consistent with the MnO₂ standard. EXAFS spectra reveal that monometallic Mn catalyst shows three major peaks at 1.5 Å, 2.4 Å and 3.1 Å, which are attributing to the Mn-O coordination, edge-sharing (Mn-Mn_{edge}) and corner-sharing (Mn-Mn_{corner}) coordination, respectively [26]. In comparison, EXAFS spectra of Fe-Mn bimetallic catalysts only show the Mn-O and Mn-Mn_{edge} coordination peaks. The slight downshift of the Mn-O coordination and upshift of the Mn-Mn_{edge} coordination indicate the mild change of the atomic distance induced by Fe. Moreover, the amplitudes of both peaks are lower than those of monometallic Mn catalyst. The absence of the Mn-Mn_{corner} shell coordination and the diminishing peak amplitude suggest a low crystallinity and small crystallite size induced by Mn. It is also noted that the signal of Mn-Mn_{edge} shell weakens as the Mn content increases, indicating that excess Mn doping enhances the amorphization of the catalysts.

3.2. Structure change in the activation process

The reducibility of the as-prepared catalysts was evaluated using H₂-TPR performed from room temperature to 900 °C (Fig. 5). It is seen that monometallic Fe catalyst shows four major reduction peaks at 355, 407, 631 and 700 °C, corresponding to reduction steps of γ -Fe₂O₃→Fe₃O₄, α -Fe₂O₃→Fe₃O₄, Fe₃O₄→FeO, and FeO→Fe⁰, respectively. Monometallic Mn catalyst shows two reduction peaks at 245 and 401 °C, corresponding to the reduction step of MnO₂→Mn₃O₄→MnO. Further reduction of MnO to metallic Mn is not observed since it can only occur at temperature higher than 1280 °C using carbon as the reducing agent [27]. The reduction profiles of FeMn(10:1) and FeMn(4:1) catalysts are similar to that of monometallic Fe, but with lower reduction temperatures, suggesting that addition of Mn in small quantities can enhance the reduction of iron oxides, which agrees with our previous findings [17]. Reduction profiles of Fe-Mn bimetallic catalysts with high Mn content are quite different from those with lower Mn content. FeMn(2:1) and FeMn(1:1) catalysts show five major reduction peaks at ca. 260, 330, 410, 500, and 680 °C, where the peaks at ca. 260 (MnO₂→Mn₃O₄) and 410 °C (Mn₃O₄→MnO) are attributing to the reduction peaks of MnO₂ on the surface layer of the catalysts. It is also seen that reduction peaks associated with the reduction steps of Fe₃O₄→FeO and FeO→Fe⁰ shift to higher temperatures (ca. 500 and 680 °C) and became more distinguishable than those in catalysts with lower Mn content, suggesting that high level of Mn retards the reduction process of FeO to metallic Fe.

The structure evolution of the three typical Fe-Mn catalysts (monometallic Fe, FeMn(4:1) and FeMn(1:1)) during reduction was further characterized with in situ XRD (Fig. 6). The diffraction peaks slightly shift towards the lower angles as the temperature increases, indicating that high temperature leads to lattice expansion. It is noted that the downshift of diffraction patterns is more recognizable in the Mn doped Fe catalysts, suggesting the incorporation of Mn into the lattice of Fe₂O₃, rendering a more temperature-sensitive lattice structure. The in situ XRD patterns of monometallic Fe catalyst confirm the reduction path of Fe₂O₃→Fe₃O₄→FeO→Fe⁰, which support the H₂-TPR results. After 5 h reduction at 350 °C, the monometallic Fe catalyst is mostly made up of metallic Fe with little FeO. The diffraction patterns of Mn doped catalysts do not only show features of metallic iron, iron oxides and manganese oxide, but also contain Fe-Mn bimetallic oxides in the form of

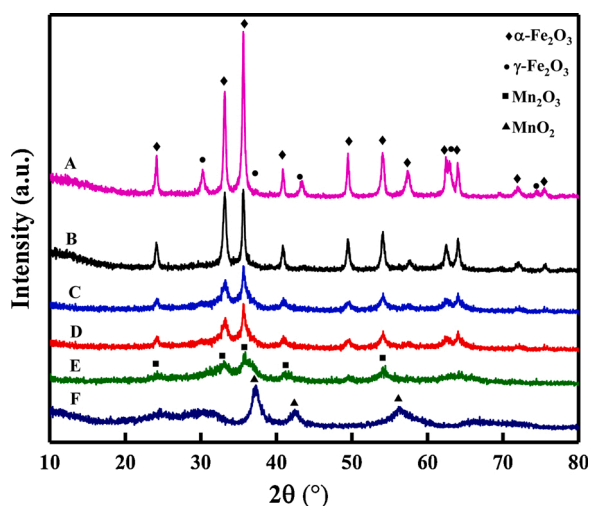


Fig. 1. XRD patterns of the as-prepared Fe-Mn bimetallic catalysts (A) Fe; (B) FeMn(10:1); (C) FeMn(4:1); (D) FeMn(2:1); (E) FeMn(1:1); (F) Mn.

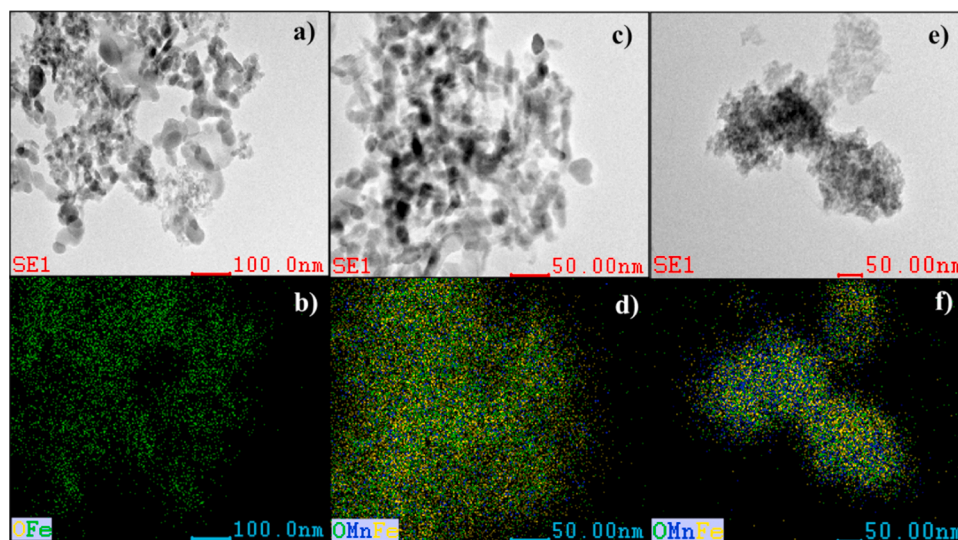


Fig. 2. TEM images and EDS mappings of the as-prepared Fe-Mn bimetallic catalysts (a-b) Fe; (c-d) FeMn(4:1); (e-f) FeMn(1:1).

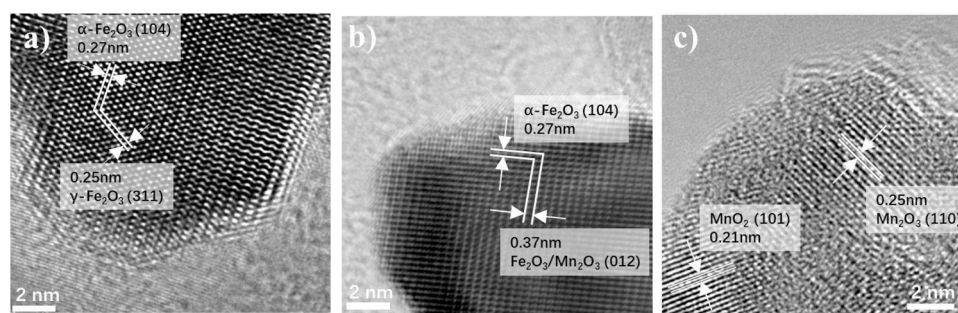


Fig. 3. HR-TEM images and line profiles of the as-prepared Fe-Mn bimetallic catalysts (a) Fe; (b) FeMn(4:1); (c) FeMn(1:1).

$\text{Fe}_{3-x}\text{Mn}_x\text{O}$ (e.g., $\text{Fe}_{1.7}\text{Mn}_{1.3}\text{O}_4$, Fe_2MnO_4 , $\text{Fe}_{1-x}\text{Mn}_x\text{O}$ or the combination of these two components (e.g., $\text{Fe}_5\text{Mn}_2\text{O}_9$). For FeMn(4:1), weak signals of metallic Fe are detected after the catalyst is reduced at 350 °C for 4 h. In contrast, the monometallic Fe catalyst shows a strong peak of metallic Fe after being reduced at 350 °C for 1 h. The difference in the occurrence and peak intensity of metallic Fe indicates that the presence of Mn retards the reduction of Fe (II) to metallic Fe. $\text{Fe}_{1-x}\text{Mn}_x\text{O}$ and $\text{Fe}_{3-x}\text{Mn}_x\text{O}_4$ are the two major phases detected in the reduced Fe-Mn catalysts. It is also seen that further increase in Mn content brings forward the reduction of $\text{Fe}_{3-x}\text{Mn}_x\text{O} \rightarrow \text{Fe}_{1-x}\text{Mn}_x\text{O}$ and postpones the reduction of $\text{Fe}_{1-x}\text{Mn}_x\text{O} \rightarrow \text{Fe}^\circ$. Based on these results, we can deduce that the addition of Mn changes the reduction behavior of Fe_2O_3 by forming two major Fe-Mn bimetallic oxides with distinct reducibility (e.g., reduction difficulty: $\text{Fe}_{1-x}\text{Mn}_x\text{O} \rightarrow \text{Fe}^\circ > \text{Fe}_{3-x}\text{Mn}_x\text{O}_4 \rightarrow \text{Fe}_{1-x}\text{Mn}_x\text{O}$), thus giving rise to a mixture of the both components in the reduced bulk catalysts. One may notice that in situ XRD shows that the addition of Mn can stabilize the Fe-Mn mixing oxide ($\text{Fe}_{1-x}\text{Mn}_x\text{O}$) phase, while H_2 -TPR profile indicates that Mn enhances the reduction of iron oxides. The similar H_2 -TPR and in situ XRD results have been also reported in our previous work on Mn-containing ferric spinel microspheres [17]. We ascribed such phenomenon to the difference in these two characterization techniques. H_2 -TPR measures the H_2 consumption, while in situ XRD examines the phase transition of the catalysts in presence of H_2 reduction condition. Mn tends to form highly stable Fe-Mn mixing oxides (reducing temperature > 1100 °C) which do not show any H_2 consumption peaks on H_2 -TPR [28]. As the Mn content increases, the content of the reducible iron oxides and catalyst grain size reduce, thus enhancing the reduction of these iron oxides, as confirmed by the downshifts of the reduction temperature. On the other hand, the diffraction pattern of Fe-Mn mixing

oxides can be clearly identified during the process of reduction, indicating that the transformation of iron species to metallic iron is suppressed by Mn. Based on the observations from H_2 -TPR and in situ XRD, it can be concluded that Mn stabilizes the Fe (II) in the form of Fe-Mn mixed oxides and suppresses the phase transition of Fe (II) to metallic iron.

The structures of the reduced catalysts were further characterized by EXAFS spectroscopy. Fig. 7(a)~(c) shows the Fe K-edge XANES spectra of the reduced catalysts as well as the reference standards. The monometallic Fe catalyst shows spectral features that resemble those of the standard Fe foil, with a strong shoulder at pre-edge (7112 eV). The XANES spectra of FeMn(4:1) and FeMn(1:1) show a weak pre-edge shoulder at 7114 eV, an absorption edge at 7119 eV and a white line at 7129 eV (Fig. 7 (b)), which are typical characteristics of cubic FeO. The results shown in R space indicate that the short-range structure of the reduced monometallic Fe catalyst is consistent with that of the Fe foil. In comparison, the reduced Fe-Mn catalysts showed a peak at 2.7 Å, corresponding to the Fe-Fe coordination shell in FeO. Interestingly, the contribution of Fe-O coordination shell is very weak in Fe-Mn catalysts. It is also seen that the peak at 2.7 Å attenuates as the Mn content increases, indicating the growth of disorder. We further analyzed the Mn K-edge XANES spectra of the reduced catalysts (Fig. 7(d)~(f)). It is seen that the Fe-Mn catalysts and monometallic Mn catalyst exhibit the similar features to standard bulk MnO, with a pre-edge at 6540 eV, an absorption edge at 6545 eV and a white line at 6555 eV. The R space results from EXAFS oscillations validate that the short-range structures of the Fe-Mn catalysts match those of MnO, which exhibits two major peaks at 1.8 Å and 2.7 Å, corresponding to the Mn-O coordination shell and the Mn-Mn/Fe coordination shell. Interestingly, the peak associated

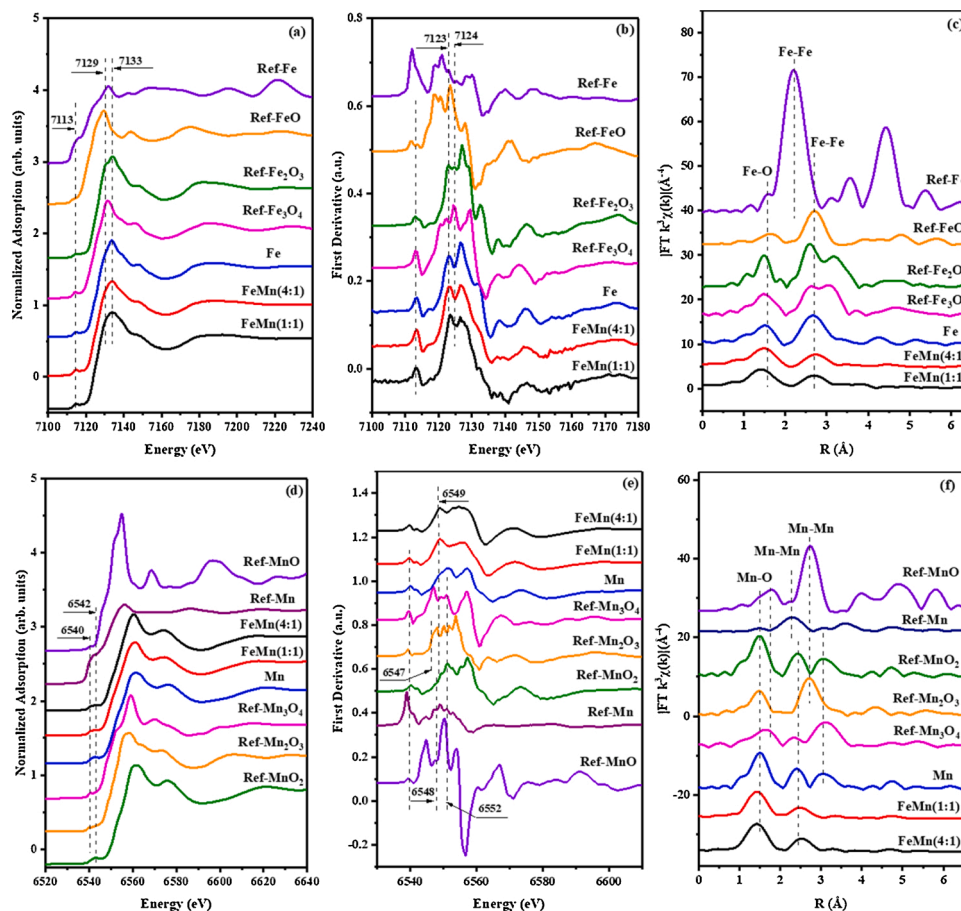


Fig. 4. K-edge XANES, first-order derivative spectra, and Fourier-transformed EXAFS spectra of the as-prepared catalysts. Fe: (a)–(c); Mn: (d)–(f).

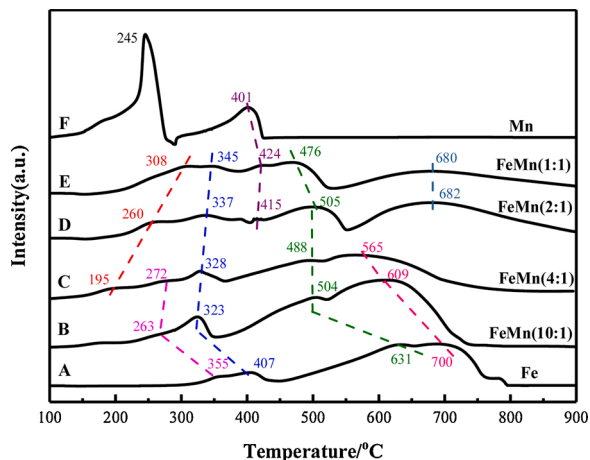


Fig. 5. H_2 -TPR profiles of Fe-Mn catalysts with different Fe/Mn ratios.

with the Mn-O shell slightly shifts to the lower distance upon Mn doping. This indicates that parts of Mn atoms are replaced by smaller Fe atoms, resulting in a shorter scattering path.

3.3. Characterization of the spent catalysts

TEM images of the spent catalysts (Fig. 8) evidence that the presence of Mn dramatically suppresses the growth of the particle size. The elemental distribution on catalyst surface given by EDS mapping (Table 2) shows that the surface layer of the spent catalysts is rich in C,

O, Fe and Mn. To circumvent the influence of the carbon film on elemental mapping, we have carefully selected the spots located in the copper grid opening which was not covered by the carbon film for imaging. The surface carbon content shows a decreasing trend whereas the surface oxygen content shows an increasing trend towards the increasing Fe/Mn ratio in the as-prepared catalysts, manifesting that Mn can suppress the coke formation on catalyst surface. Spent FeMn(4:1) catalyst shows a surface Fe/Mn atomic ratio of 8, implying the enrichment of Fe on the catalyst surface. In contrast, the spent FeMn(1:1) gives a smaller Fe/Mn atomic ratio than the nominal value, suggesting that the outer layer of the catalyst is surrounded by Mn, which matches the MnO_2 layer observed on the TEM image of the as-prepared FeMn(1:1) catalyst (Fig. 3(c)).

The phase compositions of the spent catalysts are characterized by XRD (Fig. 9). Spent Fe catalyst is made up of Fe_3O_4 and of $\chi\text{-Fe}_5\text{C}_2$. The diffraction peaks at 30.1° , 35.4° , 37.1° , 56.9° and 62.5° are characteristics of the (2 2 0), (3 1 1), (2 2 2), (5 1 1) and (4 0 0) planes of cubic Fe_3O_4 (ICDD PDF#04–009-4225). Diffraction peaks of $\chi\text{-Fe}_5\text{C}_2$ are observed at 2θ of 39.4° , 40.9° , 41.4° , 43.5° , 47.3° , 50.2° and 58.3° (ICDD PDF#00–036-1248). With the addition of Mn, the diffraction peaks of Fe_3O_4 vanish, whereas the signals of $\text{Mn}_{0.503}\text{Fe}_{0.497}\text{O}$, MnCO_3 , and $\varepsilon\text{-Fe}_2\text{C}$ appear. It is also seen that the diffraction peaks of $\text{Mn}_{0.503}\text{Fe}_{0.497}\text{O}$ and MnCO_3 become more distinguishable while the signals of $\chi\text{-Fe}_5\text{C}_2$ become weaker as the Mn content increases. No diffraction peaks of $\chi\text{-Fe}_5\text{C}_2$ are detected in the spent FeMn(1:1) catalyst. Table 3 listed the semi-quantitative phase composition and the average crystallite size of the spent catalysts determined by Rietveld refinement of the XRD patterns. The total iron carbides ($\chi\text{-Fe}_5\text{C}_2 + \varepsilon\text{-Fe}_2\text{C}$) in the bulk phase of the spent catalysts shows a decreasing trend (from 84 % to 15.6 %) as the Mn content increases, evidencing that the addition of Mn

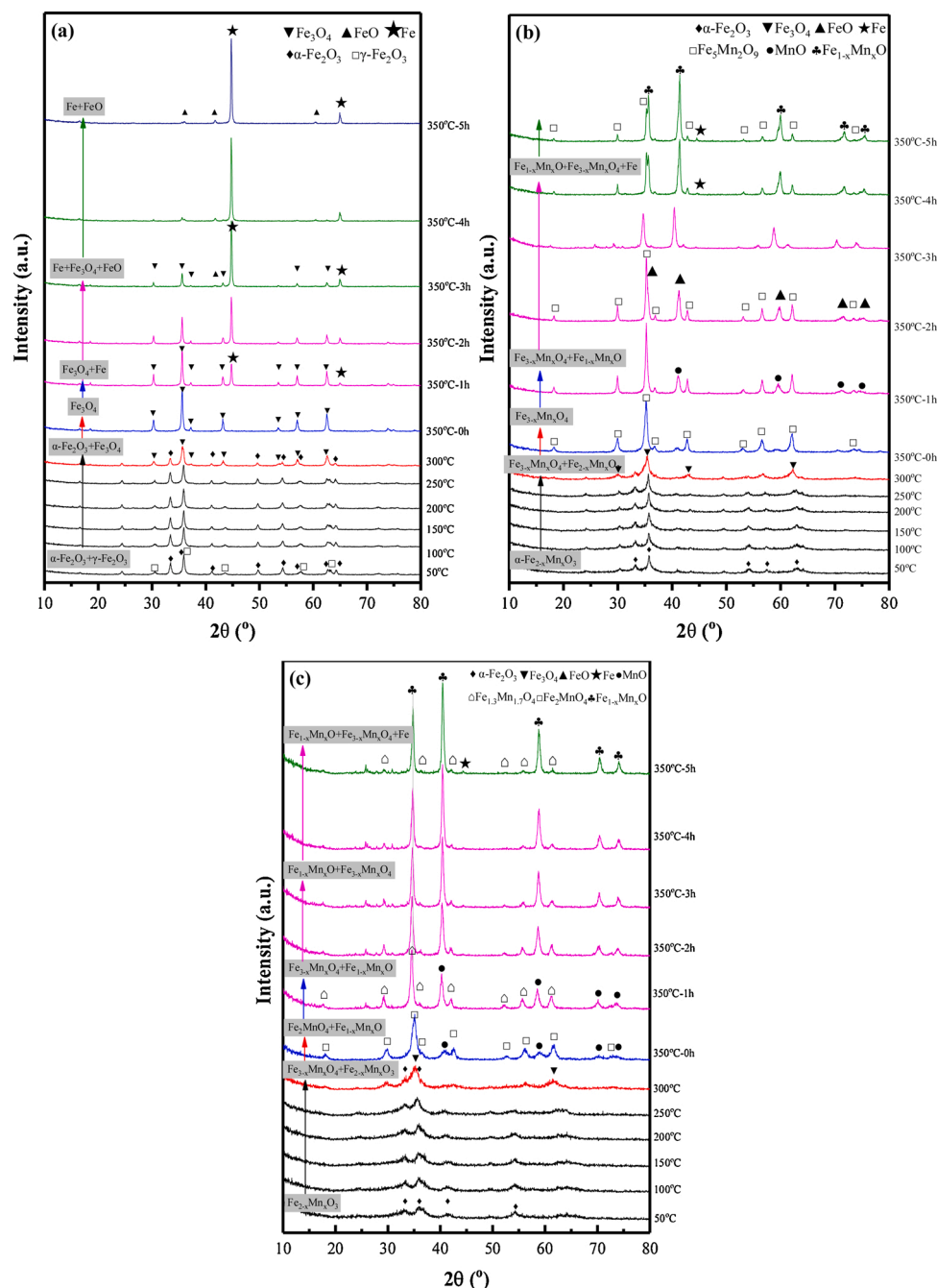


Fig. 6. In situ XRD patterns of monometallic Fe (a), FeMn(4:1) (b), FeMn(1:1) (c) catalysts.

suppresses the formation of iron carbides. It is also noted that the ratio of χ -Fe₅C₂/ε-Fe₂C decreases gradually as the Mn content increases in the catalyst. The calculated average crystallite size declines as the Mn content increases, which matches the observations in the TEM analysis, manifesting that the addition of Mn can retard the growth of the particle size during reaction.

The spent catalysts are characterized by MES at room temperature to determine the phase composition of iron carbide species. The spectra and the fitting parameters are shown in Fig. 10 and Table 4. The spent catalysts show the characteristics of iron carbides (χ -Fe₅C₂, ε'-Fe_{2.2}C, and θ-Fe₃C), metallic iron and iron oxides (superparamagnetic Fe²⁺/Fe³⁺). As the Mn content increases, the content of χ -Fe₅C₂ decreases, in accordance with the quantitative analysis of XRD result. MES results further confirm that the addition of Mn suppresses the carburization of iron oxides.

To characterize the chemical states of the catalyst before and after reaction, we have collected the Fe 2p XPS spectra of the as-prepared and spent catalysts. As shown in Fig. 11(a), the Fe 2p spectra of the as-prepared catalysts exhibit two major contributions located at 710.2 and 723.9 eV, corresponding to the spin-coupled doublet for Fe 2p_{3/2} and Fe 2p_{1/2} of Fe (III). The shoulder at 718.7 eV is assigned to the shakeup satellite of Fe (III). The results of the surface composition (Table 5) reveal that the surface Fe/Mn atomic ratio is smaller than that of the bulk catalyst, suggesting the enrichment of the Mn species on the surface of the Fe-Mn bimetallic catalysts. Due to the severe carbon deposition, no XPS signals in Fe 2p spectra are detected for the spent catalysts until being sputtered in Ar⁺ beam. The Fe 2p spectra of the spent catalysts evidence two peaks at 707.3 and 720.7 eV, corresponding to χ -Fe₅C₂ phase. The calculated (Fe + Mn)/O atomic ratios of the spent Fe, FeMn(4:1), and FeMn(1:1) are 2.01, 1.21 and 1.06, suggesting that

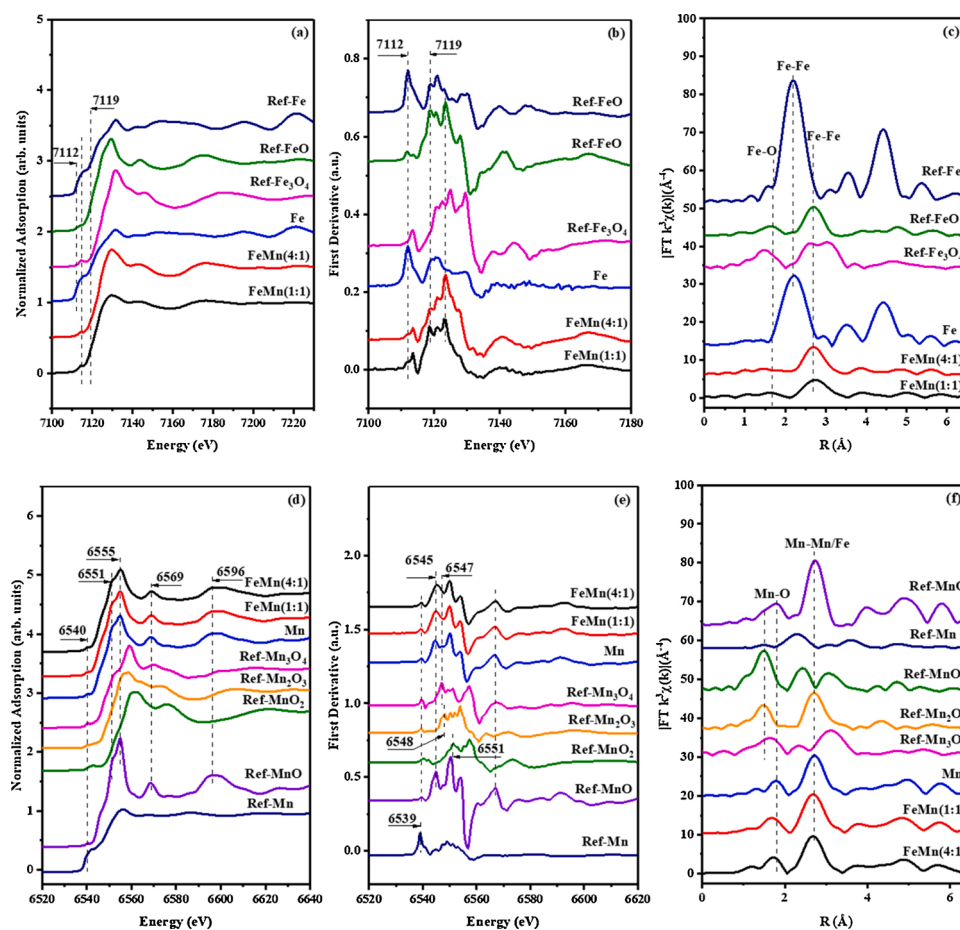


Fig. 7. K-edge XANES, first-order derivative spectra, and Fourier-transformed EXAFS spectra of the reduced catalysts. Fe: (a)~(c); Mn: (d)~(f).

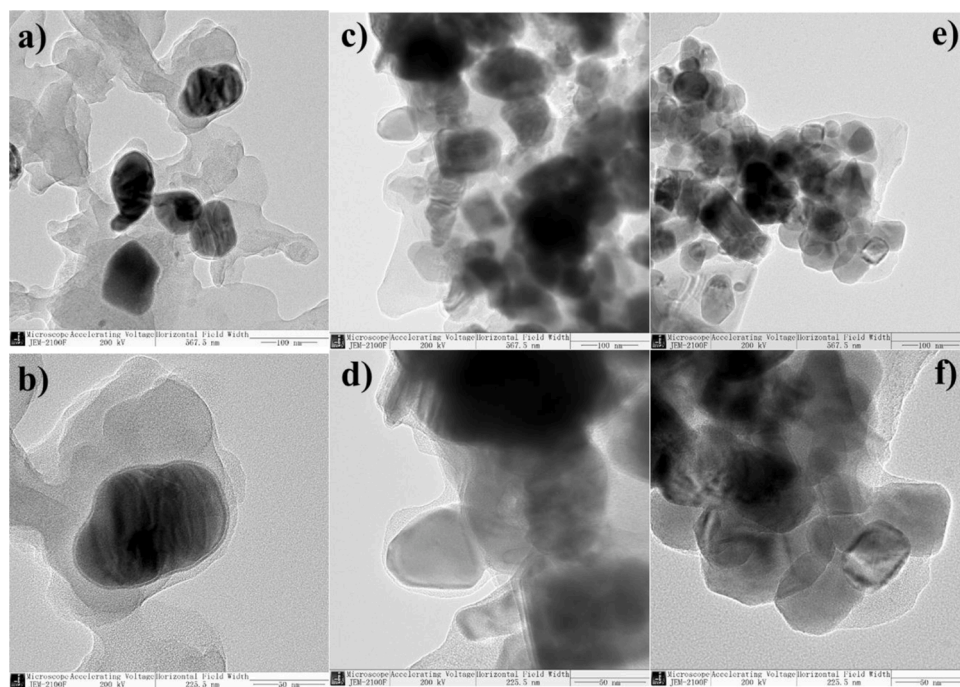
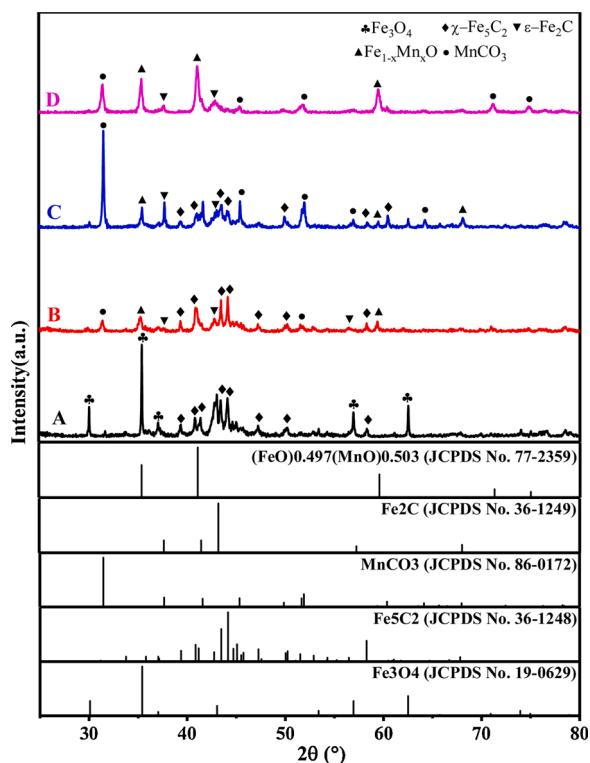


Fig. 8. TEM images of the spent catalysts, (a, b) Fe; (c, d) FeMn(4:1); (e, f) FeMn(1:1).

Table 2

Elemental composition of the spent catalysts determined by TEM-EDS.

Catalyst	Atomic percent (%)			
	C	O	Fe	Mn
Fe	83.6	3.5	12.9	N.A.
FeMn(4:1)	77.2	8.1	13.1	1.6
FeMn(1:1)	53.1	28.2	8.1	10.6

**Fig. 9.** XRD patterns of the spent catalysts. (A) Fe, (B) FeMn(4:1), (C) FeMn(2:1), (D) FeMn(1:1).

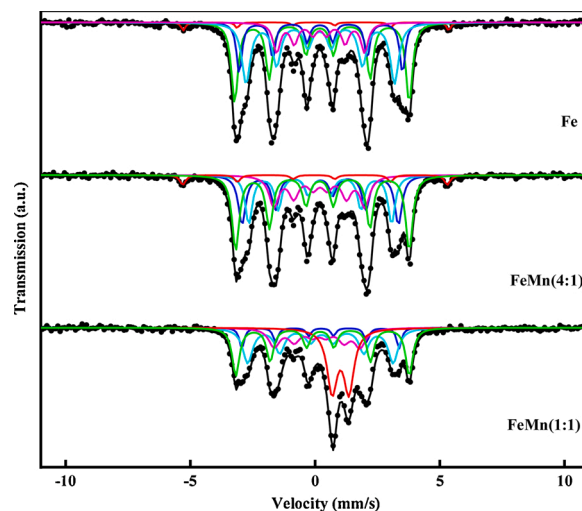
the presence of Mn retards the loss of oxygen in the bulk catalysts. Among all the three spent catalysts, the FeMn(4:1) catalyst exhibits the highest FTY, and after FTO reaction, it has the highest content of bulk carbonaceous compound (carbide, carbonate) and surface carbon content, reflecting its superior ability of CO dissociation.

XAFS techniques are used to further examine the electronic structure and the chemical bonding of the spent catalysts (Fig. 12). XANES spectra of all the spent catalysts show a pre-edge at 7112 eV, which matches the characteristic of Fe foil and iron carbides. Pure Fe and FeMn(4:1) catalysts show a white line positioned at 7131 eV, which matches the features of the Fe foil and iron carbides. As the Mn content further increases, the white line slightly shifts towards lower energy at 7129 eV, coinciding with the characteristic of FeO. Moreover, FeMn(1:1) shows a

strong post-edge signal at 7143 eV, corresponding to the multiple scattering features of FeO. Fourier-transformed EXAFS spectra show that Fe and FeMn(4:1) catalysts have the similar short-range structure with a Fe-C scattering path at 1.6 Å. However, the long-range structure of FeMn(4:1) closely resembles that of Fe foil. FeMn(1:1) shows a Fe-Fe(Mn) scattering path at 2.6 Å, which is slightly shorter than that of FeO (2.7), demonstrating that spent FeMn(1:1) catalyst has a short-range structure close to FeO. It is further confirmed that the presence of Mn tends to suppress the formation of iron carbides and stabilize the structure of Fe/Mn oxides.

XANES spectra of the Mn K-edge exhibit representative features of Mn (II) at pre-edge and absorption edge of 6539 and 6545 eV. FeMn(1:1) and pure Mn catalyst show a similar K-edge shape with MnO standard, with a white line at 6555 eV and a shoulder peak at 6551 eV. FeMn(4:1) catalyst shows a white line at 6551 eV and a post-edge peak at 6561 eV, which is ascribed to the multiple scattering of MnCO₃ [29]. These results agree with the XRD patterns of the spent catalysts (Fig. 9). The results from R-space show that the spent Mn catalyst and FeMn(1:1) have the similar short-range structure with MnO. The presence of Fe mildly changes the atomic distance, as revealed by the slight downshift of the first and second coordination peaks compared with the MnO standard. Unlike Mn and FeMn(1:1) catalysts which shows similar scattering paths with MnO standard, FeMn(4:1) shows three distinct scattering peaks (1.60, 2.53, and 3.54 Å), corresponding to the first shell of Mn-O, the second shell of Mn-Mn(Fe) and the third shell of Mn-Mn, respectively. The much lower contribution at the second shell than the first shell and the presence of the third shell indicate the coexistence of MnO and MnCO₃ in FeMn(4:1) [30].

Quantitative fitting analysis of EXAFS spectra of the spent catalysts is carried out to obtain the coordination numbers (CNs) and bond length information. The EXAFS fitting parameters and fitting spectra of Fe K-edge for various FeMn catalysts are presented in Table S3 and Fig. S4,

**Fig. 10.** Mössbauer spectra of the spent FeMn catalysts.**Table 3**

Phase composition and average crystallite size of the spent catalysts estimated by Rietveld refinement of the XRD patterns.

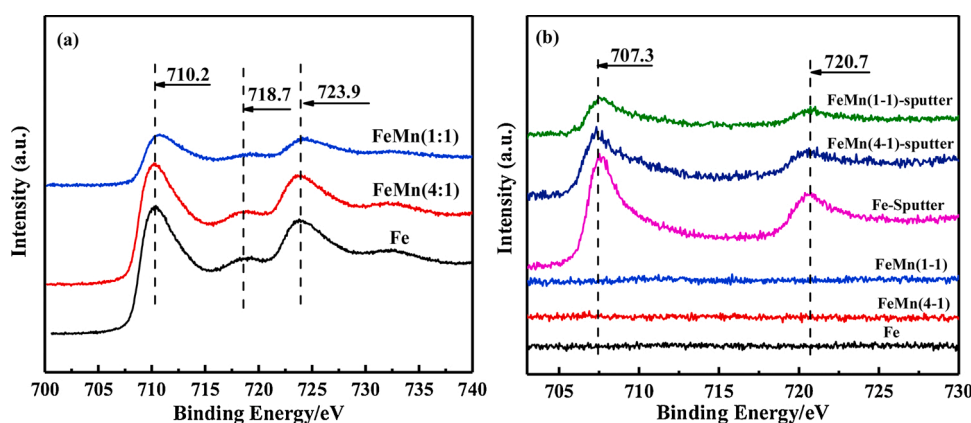
Catalyst	Phase composition (%)						Crystallite size (nm) ^c
	Fe ₃ O ₄	Mn _{0.503} Fe _{0.497} O	MnCO ₃	γ-Fe ₅ C ₂	ε-Fe ₂ C	Total carbide	
Fe	16.0 (0.9) ^a	N.D.	N.D.	84.0 (6.9)	N.D.	84.0	57.7 (4.2)
FeMn(4:1)	^b N.D.	13.3 (0.9)	9.8 (0.6)	63.4 (4.0)	13.6 (1.7)	77.0	37.9 (1.5)
FeMn(2:1)	N.D.	8.3 (0.7)	36.8 (2.1)	41.6 (3.9)	13.2 (1.2)	54.8	29.4 (3.0)
FeMn(1:1)	N.D.	47.7 (2.7)	36.7 (2.2)	N.D.	15.6 (1.5)	15.6	25.0 (0.6)

^a Standard error in the parenthesis.^b Not detected.^c Estimated from XRD patterns of all the iron phases.

Table 4

Fitting parameters of the Mössbauer spectra of the spent catalysts (corresponding to Fig. R3).

Sample	IS (mm·s ⁻¹)	QS (mm·s ⁻¹)	Hyperfine field (KOE)	Γ/2 (mm·s ⁻¹)	Area (%)	Phase	Compositions (%)
Fe	-0.01	-0.08	329.64	0.12	2	Fe ⁰	2
	0.23	-0.09	216.75	0.16	33.9	Fe ₅ C ₂ (I)	
	0.19	-0.03	184.38	0.21	33	Fe ₅ C ₂ (II)	80.4
	0.21	-0.05	109.21	0.17	13.5	Fe ₅ C ₂ (III)	
	0.21	-0.03	202.88	0.13	17.5	Fe ₃ C	17.5
FeMn(4:1)	-0.02	-0.04	328.61	0.14	3.6	Fe ⁰	3.6
	-0.11	0.24	215.55	0.17	33.7	Fe ₅ C ₂ (I)	
	0.21	0.01	194.27	0.18	22.7	Fe ₅ C ₂ (II)	74.9
	0.19	0.03	112.95	0.23	18.5	Fe ₅ C ₂ (III)	
	0.19	-0.02	177.37	0.17	21.5	Fe _{2.2} C	21.5
FeMn(1:1)	1.02	0.67		0.25	23.2	spm Fe ³⁺ /Fe ²⁺	23.2
	-0.09	0.26	215.85	0.16	24.3	Fe ₅ C ₂ (I)	
	0.03	0.23	194.54	0.18	18.2	Fe ₅ C ₂ (II)	60.2
	0.11	0.12	107.7	0.29	17.7	Fe ₅ C ₂ (III)	
	0.06	0.24	177.63	0.18	16.5	Fe _{2.2} C	16.5

**Fig. 11.** Fe2p XPS spectra of the as-prepared (a) and spent catalysts (b). (Ar⁺ sputtering conditions: 3 kV, 2 × 2 mm², 2 min).**Table 5**

Elemental distribution of the as-prepared and the spent catalysts determined by XPS.

Catalyst	Elemental distribution (At. %)			
	Fe	Mn	O	C
A-Fe ^a	20.54	N.D.	69.56	9.90
A-FeMn(4:1)	16.40	6.32	66.08	11.19
A-FeMn(1:1)	7.97	15.79	62.71	13.53
S-Fe ^b	N.D.	N.D.	3.47	96.53
S-FeMn(4:1)	N.D.	N.D.	0.71	99.29
S-FeMn(1:1)	0.09	0.31	2.53	97.07
Sputtered S-Fe	1.73	N.D.	0.86	97.41
Sputtered S-FeMn(4:1)	0.58	0.15	0.60	98.68
Sputtered S-FeMn(1:1)	1.14	1.33	2.64	94.90

^a "A" stands for "as-prepared".^b "S" stands for "spent".

respectively. The CN of the Fe-C shell in the spent Fe catalyst is 3.6, which drops to 1.9 in the spent FeMn(4:1) catalyst, corresponding to a lower content of iron carbide. The coordination shell of Fe-O is observed in the spent FeMn(4:1) catalyst, verifying that Mn retards the carburization. The CNs of the first and second shells of Fe-Fe are 4.1 and 2.8, respectively, which are much smaller than those of the spent Fe catalyst, suggesting that incorporation of Mn enhances the dispersion of iron phases, which is in accordance with the results of XRD and TEM.

3.4. Catalytic performance

The as-prepared monometallic Fe and Fe-Mn bimetallic catalysts were in situ reduced at 350 °C for 5 h and then tested in the FTS at 260 °C, 2 MPa, H₂/CO ratio of 1, and gas flow rate of 25 mL/min. The CO conversion and hydrocarbon selectivity obtained with different catalysts after 20 h on stream are displayed in Fig. 13 and Table S1. The TOS profile of the conversion and product selectivity is illustrated in Fig. S3. It is seen that the CO conversion exhibits a volcano-type relationship with the Mn content. The CO conversion exists a peak value (5.49 %) at Fe:Mn ratio of 4:1. Since Mn species are proved to be inactive for FTS, the volcano peak can be attributed to the trade-off between the decrease in the fraction of active element, which is iron in this case and the increase in the dispersion of the active element as revealed by the decreasing average crystallite size. The iron time yield (FTY) is calculated to estimate the catalytic efficiency with respect to the unit mass of iron element in the catalysts. Surprisingly, the FTY curve also shows a volcanic evolution with the Fe/Mn ratio, with a peak value of 25.5 μmol_{CO}·g_{Fe}⁻¹·s⁻¹ over Fe-Mn (4:1). We suspect that Mn does not solely promote the dispersion of iron particles but also inhibits the formation of the active iron carbides, thus giving rise to a volcano-type curve of FTY with respect to Fe/Mn ratio of the catalysts.

The product distribution of the FTS reaction over Fe-Mn catalysts with various Fe/Mn ratio is also plotted in Fig. 13. The detailed product distribution is summarized in Table S1. It is seen that monometallic Fe catalyst gives a CO₂ selectivity as high as 44 %, whereas the bimetallic Fe-Mn catalysts exhibit much lower CO₂ selectivity (20–27 %). In FTO reaction, the oxygen atom in CO can be rejected in the form of CO₂ or

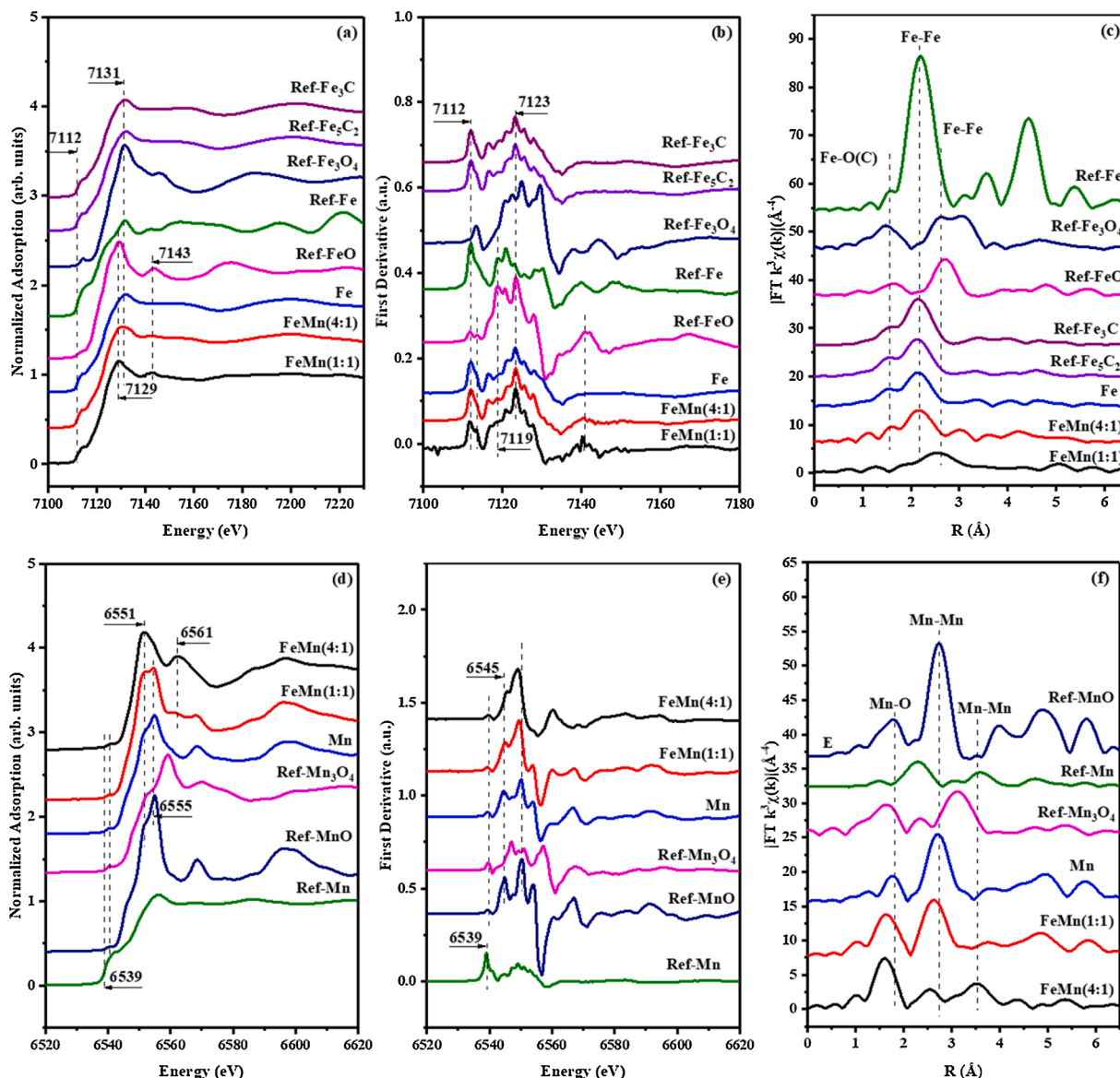
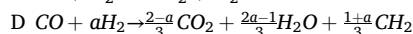
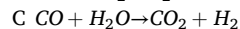
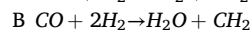
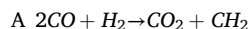


Fig. 12. K-edge XANES, first-order derivative spectra, and Fourier-transformed EXAFS spectra of the spent catalysts. Fe: (a)–(c); Mn: (d)–(f).

H₂O depending on the H₂/CO ratio in the feed (as shown in Reaction A and B). The CO₂ directly generated in FTO (reaction A) is also considered as the primary CO₂. In addition, secondary CO₂ can be produced if the oxygen atom is rejected as H₂O followed by WGS (Reaction B and C). The overall reaction that combines the contribution of FTO and WGS is given in Reaction D, which indicates that the CO₂ selectivity is varied between 0–50%, depending on various H₂/CO ratio (a , $a = 0.5\text{--}2.0$). Iglesia et al. [31] has evidenced that H-assisted CO dissociation which removes oxygen in terms of H₂O is kinetically more favorable than direct CO dissociation mechanism on Fe-based catalysts at FTO reaction conditions, pointing out that the secondary CO₂ is kinetically favorable. Considering the fact that Fe₃O₄ is the known active phase for WGS [32], the difference in CO₂ selectivity from monometallic Fe and Fe-Mn catalysts may reflect that the interaction of Mn hinders the formation of Fe₃O₄ during FTO reaction.



As for hydrocarbons, the C2–C4 olefin selectivity exhibits a volcanic evolution as the function of Fe/Mn ratio, peaking at 38.6 % with FeMn (4:1). CH₄ selectivity shows a similar variation trend to that of C2–C4 olefin, with the highest value of 15 % over FeMn(4:1). The addition of Mn tends to shift the product distribution to higher carbon number, as revealed by the variation of C5+ selectivity. Since the significant difference in CO₂ selectivity over monometallic Fe and Fe-Mn catalysts may obscure the activity of major reactions that contribute to the hydrocarbons, we have calculated the C2–C4 selectivity and chain-growth probability (α) without accounting for CO₂. The lower olefins obtained over monometallic Fe account for 46.7 % (CO₂ free) of the total hydrocarbons. As the Mn content increases, the proportion of lower olefins in the hydrocarbon product decreases after a slight increase, with a maximum of 48.7 % (CO₂ free) over FeMn(4:1). The calculated α value shows a decrease followed by an increase after reaching the minimum (0.48) over FeMn(4:1) as the Mn content increases. These results imply that the addition of Mn suppresses the chain-growth reactions to some extent and favors the termination of hydrocarbon chain in terms of lower olefins. Iron carbides are commonly accepted bulk active phases in carbon-carbon coupling reactions; therefore, it is possible that the formation of active iron carbides is retarded due to the interaction

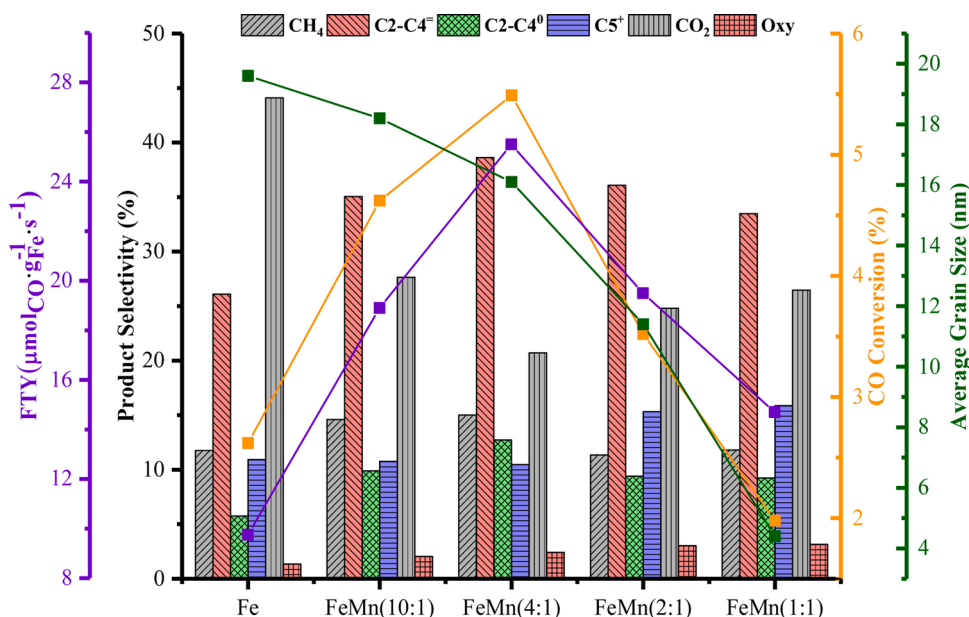


Fig. 13. Catalytic performance of the Fe-Mn catalysts with different Fe/Mn molar ratios. Reduction conditions: H₂ (99.999 vol. %), 25 mL/min, 350 °C, 0.1 MPa, 5 h. Reaction conditions: catalyst diluted with SiC (mixing ratio = 1:3), 260 °C, 2.0 MPa, H₂/CO = 1, gas flow = 25 mL/min, TOS = 20 h.

between Mn and Fe at an appropriate Mn level. Interestingly, the chain growth probability regains to the level comparable to monometallic Fe over higher Mn content (e.g., FeMn(2:1) and FeMn(1:1)), which points to the separation of bulk Fe and Mn phases at high Mn levels. FeMn catalysts have been extensively studied for FTO reaction. We have made a comparison on the catalytic performance of FeMn based catalysts prepared via common methods, such as precipitation, impregnation, solvothermal, or any combinations of these methods (Table S2). Among those unsupported FeMn catalysts, FeMn prepared by low-temperature precipitation method in this study exhibits higher FTY, which can be ascribed to the higher surface area and better dispersion of active sites. It is also noteworthy that the catalyst prepared by the low-temperature precipitation yields the lowest CO₂ selectivity.

4. Discussions

4.1. Effects of Mn on the phase structure of the as-prepared catalysts

Mn doping is an effective way to regulate the structural and chemical properties of the Fe₂O₃. During preparation, the nanocrystals tend to agglomerate to minimize the surface energy. As evidenced by XRD patterns and TEM images of the as-prepared catalysts (Figs. 1–3), the crystallinity and the particle size clearly demonstrate negative variation trends towards the Mn doping content. Due to the fact that Mn has a lower surface energy than Fe (Mn: 1.54 J m⁻², Fe: 2.42 J m⁻²), the addition of Mn can reduce the surface energy of the nanocrystals and restrain them from aggregation [33]. HR-TEM images reveal that the lattice spacing of the Fe₂O₃ increases (0.27 nm→0.37 nm) after Mn doping, pointing to the formation of solid solution of Fe/Mn oxide at iron oxide layers. The mismatch of ionic radius of Mn³⁺ (0.72 Å for low spin, 0.785 Å for high spin) and Fe³⁺ (0.64 Å) is responsible for the lattice distortion in the Fe-Mn bimetallic catalysts [34]. This is further confirmed by the r-space information, showing that Mn slightly extends the Fe-Fe bond distance while shortens the Fe-O bond distance (Fig. 4 (c)). Due to the low surface energy, MnO_x tends to migrate outwards and form a coating on the catalyst surface. This is corroborated by the fact that diffraction patterns of MnO₂ are not visible while lattice fringes of MnO₂ are shown on the surface of the FeMn(1:1) catalyst. The phase separation of manganese oxides and iron oxides have been well-documented in other studies [23,35].

4.2. Effects of Mn on the phase transition during activation and reaction

It is learnt from H₂-TPR that small amount of Mn favors the reduction of Fe₂O₃ to metallic Fe, but high level of Mn makes the reduction of Fe (II) to metallic Fe difficult again. In situ-XRD verifies the formation of stable Fe_{1-x}Mn_xO phase with poor reducibility, suggesting that the presence of Mn as a promoter suppresses the reduction of Fe (II) to metallic Fe. The results from XAFS also verify that the short-range structure of the reduced Fe-Mn bimetallic catalysts closely resembles to FeO. Tao et al. [36] pointed out that the Fe-Mn interaction enhanced the incorporation of Mn²⁺ into the FeO lattice and improved its stability. In addition, the formation of highly stable MnO coating on the surface of FeO can defer the reduction of FeO phase [37].

According to the XRD results of the spent catalysts, Fe-Mn bimetallic catalysts consist of Fe_{1-x}Mn_xO, χ -Fe₅C₂, ϵ -Fe₂C, and MnCO₃. The fraction of the active iron carbides (χ -Fe₅C₂ and ϵ -Fe₂C) decreases as the Mn content increases, manifesting that the Fe-Mn interaction further restrains the carburization process. Further XANES spectra of the Fe K-edge of the spent Fe-Mn catalysts indicate that absorption edge occurs at energies between the Fe foil and the FeO standard, suggesting that Fe is still in the oxide form after reaction. Using TPR-XANES/EXAFS technique, Ribeiro et al. [38] confirmed that the presence of high level Mn significantly hindered the carburization rates. In addition, the mixed Fe/Mn oxide phase tends to retard crystallite growth and thus, smaller iron carbide particles are retained during reaction (Table 4).

4.3. Relationship between the catalyst phase structure and catalytic performance

FTO is proved to be size-dependent and structure-dependent, which means that the activity and product selectivity is dictated by the combinational effects of particle size, phase structures and compositions of the catalyst. According to the discussion above, Mn obviously exerts an impeditive effect on the agglomeration of Fe₂O₃ nanocrystals during catalyst preparation and activation processes, giving rise to highly dispersed nanoparticles. Considering that the particle size of the spent catalysts also shows a declination towards the increasing Mn content, it is inferred that Mn acts as a structural promoter to retain a high surface area of iron carbides. On the other hand, the Fe-Mn interaction stabilizes the Fe_{1-x}Mn_xO phase and results in a poor degree of carburization. Such

trade-off between the gain in dispersion and loss in the content of active phases is clearly reflected as the volcano-style activity (Fig. 13). The contributions from different iron carbide phases on the catalytic activity have been comprehensively studied by Chang et al. [39]. Among the three carbides (ϵ -Fe₂C, Fe₇C₃, and χ -Fe₅C₂), Fe₇C₃ showed the highest intrinsic activity (TOF = $4.59 \times 10^{-2} \text{ s}^{-1}$), followed by χ -Fe₅C₂ (TOF = $1.62 \times 10^{-2} \text{ s}^{-1}$) and ϵ -Fe₂C (TOF = $1.50 \times 10^{-2} \text{ s}^{-1}$) in typical medium temperature FTO conditions (260–300 °C, 2–3 MPa, and H₂/CO = 2) [39]. As predicted by chemical carbon potential (μ_C) [40], the carbide phase evolved in a sequence of ϵ -Fe₂C/ ϵ' -Fe_{2.2}C → χ -Fe₅C₂ → θ -Fe₃C, where χ -Fe₅C₂ is known to be most active phase and carbon-deficient θ -Fe₃C phase signifies the deactivation. The increasing ratio of ϵ -Fe₂C/ χ -Fe₅C₂ as a function of Mn content manifests that Mn extends the phase transition of the iron carbide to χ -Fe₅C₂, and therefore decreases the catalytic activity.

The promotional effect of Mn on the surface basicity has been extensively studied [14,36]. It is reported that the improvements on surface basicity facilitate the CO dissociation, improves the selectivity towards olefins and restrains secondary hydrogenation of olefins. The improvement on CO dissociation by Mn addition is evidenced by the decrease in the onset temperature of the desorption of CH₄, C₂H₄ and CO₂ in syngas-TPSR (Fig. S5.). Regarding the product distribution, Mn-promoted Fe catalysts show drastically lower CO₂ selectivity than pure Fe catalyst. Since CO₂ is mainly contributed by Fe₃O₄-catalyzed WGS, the decrease in CO₂ selectivity infers the resistance of iron carbide phases to oxidation in Mn-promoted Fe catalysts. The hydrocarbon selectivity is reported to be governed by the particle size and the features of the iron carbide phases [41]. It is considered that χ -Fe₅C₂ forms larger particles than ϵ -Fe₂C, and the latter favors the C5+ selectivity while suppresses the formation of methane [42]. In addition, the difference in selectivity can be resulted from the difference in the intrinsic behaviors of the iron carbides. Many studies have testified that ϵ -Fe₂C has superior C-C coupling ability and χ -Fe₅C₂ exhibits the strongest CO dissociation ability among all the iron carbides [43,44]. Therefore, the changes in the ratio of these two carbides may explain the increase in C5+ selectivity and decrease in C2-C4 olefins selectivity as a function of Mn content. In the present study, we have noticed an increase in C5+ selectivity at high

Mn levels, which can be explained as the decrease in particle size and the increase in ϵ -Fe₂C content in the phase composition. High C2-C4 olefins selectivity is favored by strong CO dissociation and modest chain growth ability, and these two properties can be facily manipulated by optimizing the Mn content. It is also noted that MnCO₃ is detected in the spent catalysts (Fig. 9). Since MnCO₃ phase is inactive for FTO reaction, its enrichment in the bulk catalyst may block the access to the active iron carbides, which can further lead to an activity decrease.

4.4. Improvements on catalytic performance by varying activation conditions

Activation pretreatment plays an essential role in the formation of the active iron carbide species. Thus, we have further investigated the influence of activation agents (e.g., syngas, H₂ and CO) on FTO performance over the best-performed FeMn(4:1) catalyst. As shown in Fig. 14, the catalyst activated with CO and syngas demonstrates higher CO conversion (7.6 % and 7.7 %) than that activated with H₂ (4 %). It should be clarified that the H₂ concentration used for study of reducing agents is much lower than that used for catalyst screening, thus resulting in different reduction degree of the same catalyst. With respect to hydrocarbon distribution, syngas-activated catalyst exhibited the highest C2-C4 olefins selectivity (60.6 %, CO₂ free), modest CH₄ selectivity (20.3 %, CO₂ free) and the lowest C5+ selectivity (4.4 %, CO₂ free). In comparison, CO-activated and H₂-activated catalysts showed the highest C5+ selectivity (13.5 %, CO₂ free) and the highest CH₄ selectivity (22.0 %, CO₂ free), respectively. The formation of active iron carbides on catalyst layers during activation plays a critical role in determining the FTO performance. Activation in H₂ may result in a phase structure that is dominated by metallic iron, which is not active in FTO reaction. Therefore, an extra step for transformation of metallic iron to iron carbides is needed during the induction period upon exposure to syngas. When activated by syngas or CO, the iron oxide is rapidly converted into iron carbides, leading to high activity in FTO.

The activity of the catalysts is closely related to the phase structure and particle size of the active iron carbides formed during reduction and carburization process. Various types of iron carbides (such as χ -Fe₅C₂,

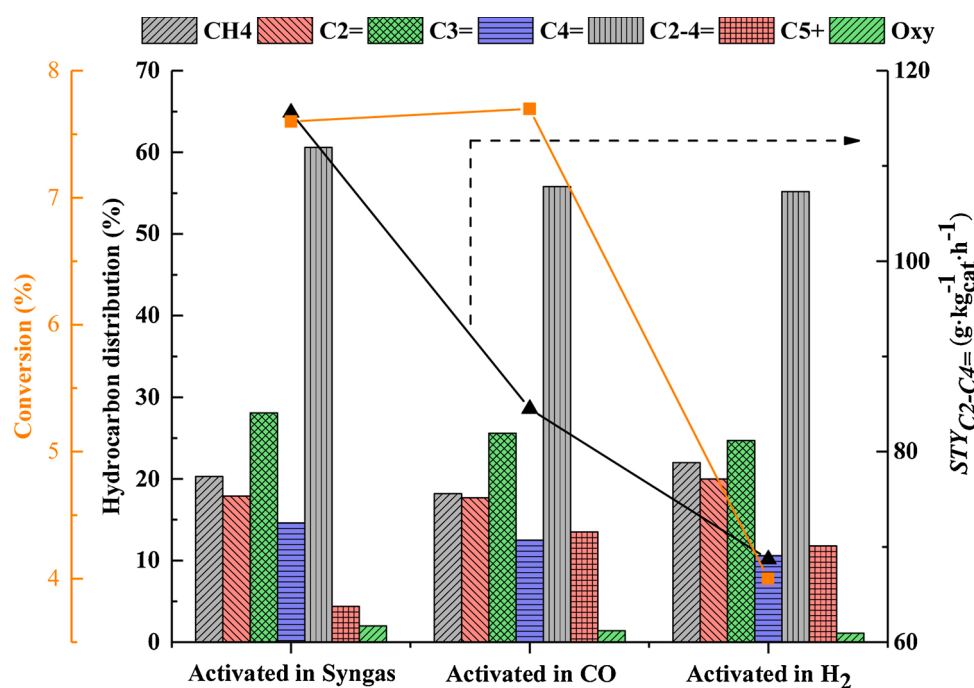


Fig. 14. Catalytic performance of FeMn(4:1) catalyst activated with different agents. Reduction conditions: Syngas (5%CO/5%H₂/90 %Ar), H₂ (5%H₂/95 %Ar), and CO (5%CO/95 %Ar), 25 mL/min, 350 °C, 0.1 MPa, 5 h. Reaction conditions: 100 mg catalyst, 260 °C, 2.0 MPa, H₂/CO = 1, gas flow = 25 mL/min, TOS = 20 h.

ε -Fe₂C, ε' -Fe_{2.2}C, Fe₇C₃ and θ -Fe₃C have been identified during the reduction and carburization process of iron oxides [40]. Different features of iron carbides have been reported to exhibit different activity and product selectivity in FTO. For instance, χ -Fe₅C₂, ε -Fe₂C, ε' -Fe_{2.2}C and Fe₇C₃ are known as active in FTO, while θ -Fe₃C is usually considered as an inactive phase since it is commonly found in the deactivated catalyst [39]. It is also reported that ε -Fe₂C has superior C-C coupling ability and enables the formation of C₅+ olefins, while χ -Fe₅C₂ is well-known for selectively producing C₂-C₄ olefins [45]. Besides the formation of various phase structures of iron carbides with distinct intrinsic activities, the FTO reaction is reported to be size-dependent in terms of both activity and selectivity [46,47]. Based on the analysis above, reducing iron oxides with different reducing agents can result in variations in carburization degree, phase structures and the particle size. In the present study, the poor activity of H₂-activated catalyst may mainly attribute to its low carburization degree, which is in line with previous reports in literature [48,49]. Product selectivity discrepancy between syngas-activated and CO-activated catalysts could be more likely related to the difference in phase compositions or particle size, but is unlikely due to carburization degree since they both show very close activity. Nevertheless, further investigation is needed to clarify the causes resulting such difference. Catalytic stability test of FeMn(4:1) implies that FeMn(4:1) catalyst maintained stable for 70 h TOS at conversion levels of 5–6 %, with a C₂-C₄ olefin selectivity over 48 % (Fig. S6).

5. Conclusions

The FTO activity and selectivity of a series of Mn-doped bimetallic bulk Fe catalysts with different Mn content were studied. The particle size, phase structures and phase content of the catalysts have been examined by TEM, ex situ/in situ XRD, UV-vis, XAFS, and XPS. It is observed that Mn can incorporate into the framework of Fe₂O₃ during catalyst calcination. The mismatch of ionic radius of Mn and Fe leads to the lattice expansion, which further causes the decrease in crystallinity and the particle size. Investigations from in situ- XRD and H₂-TPR reveal that Mn drastically stabilizes the FeO phase by forming a mixture of Fe-Mn oxide (Fe_{1-x}Mn_xO) and retards the reduction period of FeO→Fe⁰, which results in a poor reduction/carburization degree. The trade-off between the improvements on dispersion of the nanoparticles and the loss in the active iron carbides jointly contributes to the FTO activity. The superior stability of Fe_{1-x}Mn_xO endows excellent resistance towards carburization, thus giving rise to different phase compositions of the iron carbides during reaction. In addition, Mn promotes the surface basicity of the catalyst, which presents excellent CO dissociation and resistance towards secondary hydrogenation of olefins. The combination effects of the catalyst surface basicity and the phase composition of the iron carbide resulted in an appropriate Mn level for lower olefin production. Finally, a volcano-type activity versus Fe/Mn ratio curve was obtained, with a maximum CO conversion (5.49 %) over FeMn(4:1). Besides, a maximum C₂-C₄ olefin selectivity (60.6 %, CO₂ free) is achieved with FeMn(4:1) catalyst activated with syngas.

CRediT authorship contribution statement

Zixu Yang: Conceptualization, Methodology, Validation, Formal analysis, Writing - original draft, Writing - review & editing. **Zhengpai Zhang:** Conceptualization, Methodology, Investigation, Writing - review & editing. **Yitao Liu:** Methodology, Formal analysis. **Xiaoxu Ding:** Visualization, Validation. **Jun Zhang:** Visualization, Validation. **Jing Xu:** Project administration, Resources, Supervision, Funding acquisition, Writing - review & editing. **Yifan Han:** Resources.

Declaration of Competing Interest

The authors report no declarations of interest.

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Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:<https://doi.org/10.1016/j.apcatb.2020.119815>.

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